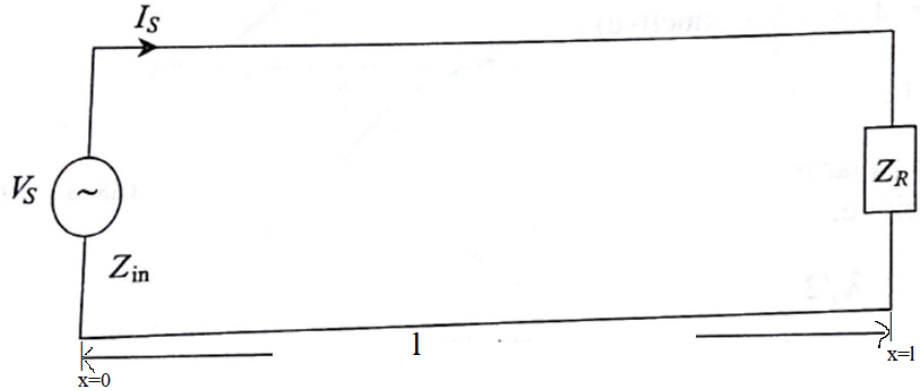
UNIT-V

Transmission Lines - II

Input Impedance:

A transmission line terminated with any load impedance Z_R at x=l is shown in fig.



- Fig . Input impedance of transmission line

The V and I equations at a point of distance x from the sending end in terms of source voltage and current are given by.

$$V = V_s \cosh \gamma x - I_s Z_0 \sinh \gamma x$$

$$I = I_s \cosh \gamma x - \frac{V_s}{Z_0} \sinh \gamma x$$

The voltage and currents at load end (x=l) are given by

$$V_R = V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l$$

$$I_R = I_s \cosh \gamma l - \frac{V_s}{Z_0} \sinh \gamma l$$

The load impedance Z_R is given by

$$Z_R = \frac{V_R}{I_R}$$

$$Z_R = \frac{V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l}{I_s \cosh \gamma l - \frac{V_s}{Z_0} \sinh \gamma l}$$

$$Z_R \left(I_s \cosh \gamma l - \frac{V_s}{Z_0} \sinh \gamma l \right) = V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l$$

$$Z_R I_s \cosh \gamma l - Z_R \frac{V_s}{Z_0} \sinh \gamma l = V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l$$

$$Z_R I_s \cosh \gamma l + I_s Z_0 \sinh \gamma l = V_s \cosh \gamma l + Z_R \frac{V_s}{Z_0} \sinh \gamma l$$

$$I_s (Z_R \cosh \gamma l + Z_0 \sinh \gamma l) = V_s \left(\cosh \gamma l + \frac{Z_R}{Z_0} \sinh \gamma l \right)$$

$$\frac{V_s}{I_s} = \frac{Z_R \cosh \gamma l + Z_0 \sinh \gamma l}{\cosh \gamma l + \frac{Z_R}{Z_0} \sinh \gamma l}$$

The input impedance of the line is given by

$$Z_{in} = \frac{V_s}{I_s}$$

$$Z_{in} = Z_0 \left[\frac{Z_R \cosh \gamma l + Z_0 \sinh \gamma l}{Z_0 \cosh \gamma l + Z_R \sinh \gamma l} \right]$$

$$Z_{in} = Z_0 \left[\frac{Z_R + Z_0 \tanh \gamma l}{Z_0 + Z_R \tanh \gamma l}\right]$$

For lossless line γ=jβ

$$Z_{in} = Z_0 \left[\frac{Z_R + jZ_0 \tan \beta l}{Z_0 + jZ_R \tan \beta l}\right]$$

<u>Voltage and Current at any point on the transmission line:</u>

Consider the transmission line of length *l* terminating with an impedance Z_R. Let V_R and I_R be the voltage and current at the load Z_R.

The voltage and current at a point of distance x from the sending end in terms of source voltage and current are given by

$$V = A \cosh \gamma x + B \sinh \gamma x \quad --- (1)$$

$$I = -\frac{1}{Z_0}(A \sinh \gamma x + B \cosh \gamma x) \quad --- (2)$$

Where A and B are constants.

At x=*l* V= V_R and I=I_R

Substituting these values in eq.(1) and eq.(2)

$$V_R = A \cosh \gamma l + B \sinh \gamma l \quad ----(3)$$

$$I_R = -\frac{1}{Z_0}(A \sinh \gamma l + B \cosh \gamma l) \quad --- (4)$$

Multiplying Eq.(3) with coshγl and Eq.(4) with sinhγl

$$V_R \cosh \gamma l = A \cosh^2 \gamma l + B \sinh \gamma l \cosh \gamma l \quad ----(5)$$

$$Z_0 I_R \sinh \gamma l = -A \sinh^2 \gamma l - B \cosh \gamma l \sinh \gamma l \quad --- (6)$$

Adding Eq.(5) and Eq.(6)

$$V_R \cosh \gamma l + Z_0 I_R \sinh \gamma l = A(\cosh^2 \gamma l - \sinh^2 \gamma l)$$

Therefore

$$A = V_R \cosh \gamma l + Z_0 I_R \sinh \gamma l$$

Multiplying Eq.(3) with sinhγl and Eq.(4) with coshγl

$$V_R \sinh \gamma l = A \cosh \gamma l \sinh \gamma l + B \sinh^2 \gamma l \quad ----(7)$$

$$Z_0 I_R \cosh \gamma l = -A \sinh \gamma l \cosh \gamma l - B \cosh^2 \gamma l \quad --- (8)$$

Adding Eq.(7) and Eq.(8)

$$V_R \sinh \gamma l + Z_0 I_R \cosh \gamma l = -(\cosh^2 \gamma l - \sinh^2 \gamma l)$$

$$B = -(V_R \sinh \gamma l + Z_0 I_R \cosh \gamma l)$$

Substituting the values of A and B in Eq.(1)

$$V = (V_R \cosh \gamma l + Z_0 I_R \sinh \gamma l) \cosh \gamma x - (V_R \sinh \gamma l + Z_0 I_R \cosh \gamma l) \sinh \gamma x$$

$$V = (V_R \cosh \gamma l \cosh \gamma x + Z_0 I_R \sinh \gamma l \cosh \gamma x) - (V_R \sinh \gamma l \sinh \gamma x + Z_0 I_R \cosh \gamma l \sinh \gamma x)$$

$$V = V_R(\cosh \gamma l \cosh \gamma x - \sinh \gamma l \sinh \gamma x) + Z_0 I_R(\sinh \gamma l \cosh \gamma x - \cosh \gamma l \sinh \gamma x)$$

$$V = V_R \cosh \gamma(l - x) + Z_0 I_R \sinh \gamma(l - x)$$

Similarly Substituting the values of A and B in Eq.(2)

$$I = I_R \cosh \gamma(l - x) + + \frac{V_R}{Z_0} \sinh \gamma(l - x)$$

If y=*l*-x then

$$V = V_R \cosh \gamma y + Z_0 I_R \sinh \gamma y$$

$$I = I_R \cosh \gamma y + + \frac{V_R}{Z_0} \sinh \gamma y$$

These are the voltage and current equations at a point of distance y from the load end in terms of terminal voltage and current.

<u>Transmission line terminated with characteristic impedance:</u>

Consider a line of length *l* which is terminated with characteristic impedance Z_0 as shown in fig.

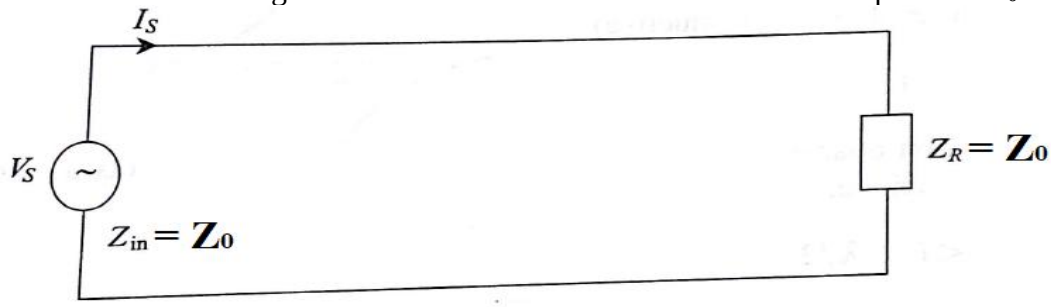


Fig. Transmission line terminated with Z_0

The input impedance of the line is given by

$$Z_{in} = Z_0 \left[\frac{Z_R + Z_0 \tanh \gamma l}{Z_0 + Z_R \tanh \gamma l}\right]$$

Here $Z_R = Z_0$

$$Z_{in} = Z_0 \left[\frac{Z_0 + Z_0 \tanh \gamma l}{Z_0 + Z_0 \tanh \gamma l}\right]$$

$$Z_{in} = Z_0$$

Hence the input impedance of a finite line terminated with characteristic impedance Z_0 is equal to characteristic impedance Z_0.

Therefore a finite line terminated with characteristic impedance Z_0 is equal to infinite line.

A line terminated with characteristic impedance Z_0 is called matched line.

<u>Standing Waves In Transmission Lines</u>: Signal energy is transmitted through a transmission line from the source to the load in the form of voltage and current waves. When the terminated load impedance is different from the characteristic impedance of the line, then the some part of the transmitted signal is returns back and there exists a reflected wave .These incident and reflected travelling waves create a standing waves.

The voltage and current equations along the transmission line equation are given by

$$V = V_i e^{-\gamma x} + V_r e^{\gamma x}$$
$$I = I_i e^{-\gamma x} + I_r e^{\gamma x}$$

In the above equations 1st term is called incident wave and 2nd term is called reflected wave.

<u>Reflection Coefficient:</u>

The reflection coefficient is defined as the ratio of reflected voltage to incident voltage. The reflection coefficient is also defined as the ratio of reflected current to incident current. It is denoted by K.

Let V_i and V_r are the incident and reflected voltages respectively.

$$K = \frac{V_r}{V_i}$$

Let I_i and I_r are the incident and reflected currents respectively.

$$K = -\frac{I_r}{I_i}$$

The voltage at the load Z_R is given by

$$V_R = V_i + V_r \quad ---(1)$$
$$I_R = I_i + I_r \quad ---(2)$$

But characteristic impedance Z_0 is $\quad Z_0 = \frac{V_i}{I_i} = -\frac{V_r}{I_i} \quad ----(3)$

Substitute the Eq.(3) in Eq.(2), we get

$$I_R = \frac{V_i}{Z_0} - \frac{V_r}{Z_0}$$

$$Z_0 I_R = V_i - V_r \quad ---(4)$$

Divide the Eq.(1) by Eq.(4), we get

$$\frac{V_R}{Z_0 I_R} = \frac{V_i + V_r}{V_i - V_r}$$

We know that

$$Z_R = \frac{V_R}{I_R}$$

$$\frac{Z_R}{Z_0} = \frac{1 + \frac{V_r}{V_i}}{1 - \frac{V_r}{V_i}}$$

We know that the reflection coefficient K is

$$K = \frac{V_r}{V_i}$$

$$\frac{Z_R}{Z_0} = \frac{1 + K}{1 - K}$$

$$Z_R(1 - K) = Z_0(1 + K)$$
$$(Z_R - Z_R K) = (Z_0 + Z_0 K)$$
$$Z_R - Z_0 = Z_R K + Z_0 K$$
$$K(Z_R + Z_0) = Z_R - Z_0$$

The reflection coefficient K is

$$\boxed{K = \frac{Z_R - Z_0}{Z_R + Z_0}}$$

The range of reflection coefficient K is $-1 \leq |K| \leq 1$

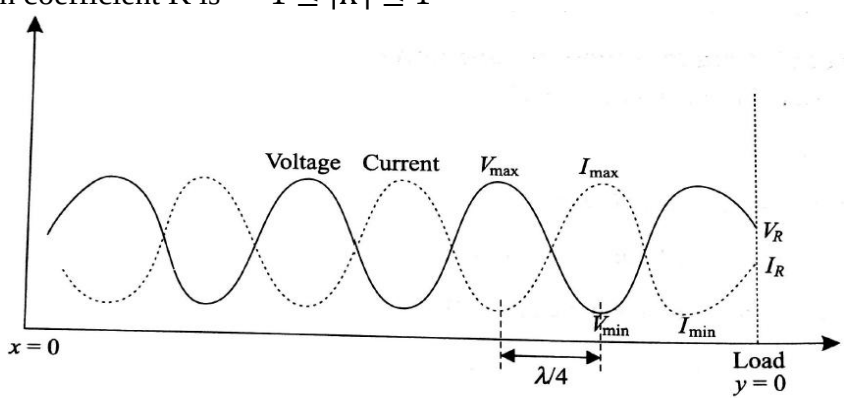


Fig. Standing wave pattern on transmission line

<u>Voltage standing wave ratio (VSWR):</u> The VSWR is defined as the ratio of the maximum voltage to the minimum voltage on the line having standing waves.

$$VSWR = S = \frac{V_{max}}{V_{min}} = \frac{|V_i| + |V_r|}{|V_i| - |V_r|} = \frac{1 + |K|}{1 - |K|}$$

$$\boxed{S = \frac{1 + |K|}{1 - |K|}}$$

Similarly the current standing wave ratio can be defined as the ratio between maximum current to the minimum current.

The range of VSWR is $1 \leq S \leq \infty$

$$(Z_{in})_{max} = \frac{V_{max}}{I_{min}} = SZ_o$$

$$(Z_{in})_{min} = \frac{V_{min}}{I_{max}} = \frac{Z_o}{S}$$

Short circuited line ($Z_R = 0$): If transmission line is terminated with short circuit, then the line is called short circuited line.

The input impedance of the short circuited line can be obtained by substituting the $Z_R = 0$ in the input impedance of the transmission line.

$$Z_{in} = Z_0 \left[\frac{Z_R + Z_0 \tanh \gamma l}{Z_0 + Z_R \tanh \gamma l}\right]$$

If Z_R=0 then

$$Z_{sc} = Z_0 \tanh \gamma l$$

The input impedance of the short circuited lossless transmission line is given by

$$Z_{sc} = jZ_0 \tan \beta l$$

The reflection coefficient of the short circuited transmission line is K= -1.

The VSWR of the short circuited transmission line is S=∞.

The variation of input impedance of SC line is shown in fig.

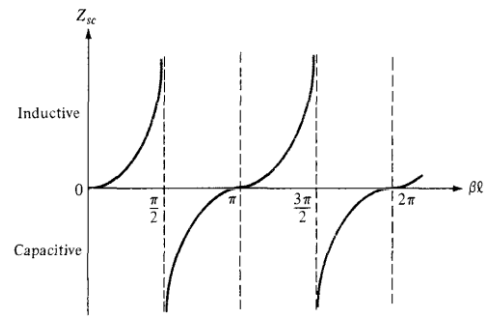


Fig . Input impedance of SC

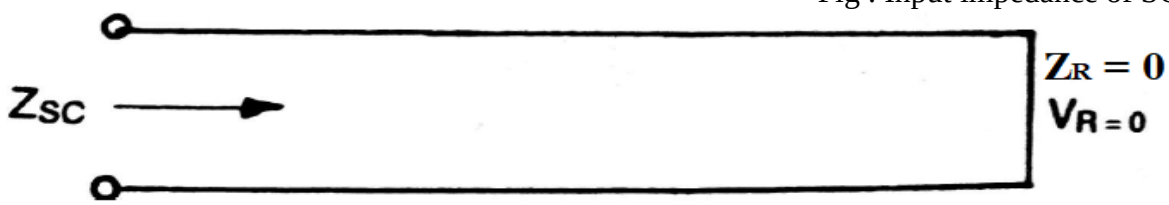


Fig. Short circuit transmission line

Open circuited line ($Z_R = \infty$): If transmission line is terminated with open circuit, then the line is called open circuited line.

The input impedance of the open circuited line can be obtained by substituting the $Z_R = \infty$ in the input impedance of the transmission line.

$$Z_{in} = Z_0 \left[\frac{Z_R + Z_0 \tanh \gamma l}{Z_0 + Z_R \tanh \gamma l}\right]$$

If Z_R=∞ then

$$Z_{oc} = Z_0 \coth \gamma l$$

The input impedance of the open circuited line lossless transmission line is given by

$Z_{oc} = -jZ_0 \cot \beta l$

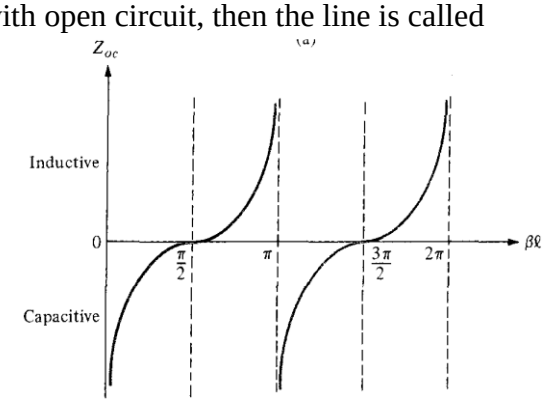


Fig. Input impedance of OC

The reflection coefficient of the open circuited line transmission line is K= 1. The VSWR of the open circuited line transmission line is S=∞. The variation of input impedance of OC line is shown in fig.

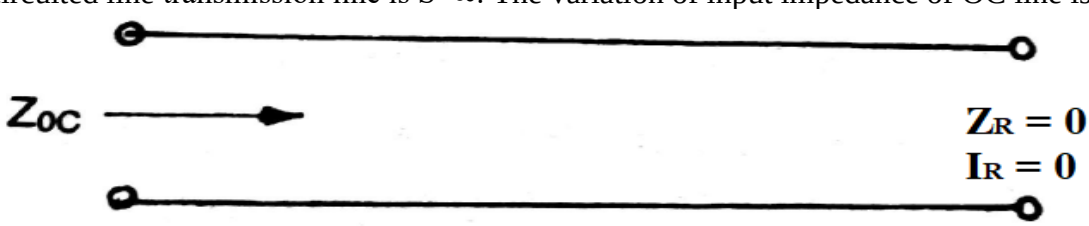


Fig. Open circuit transmission line

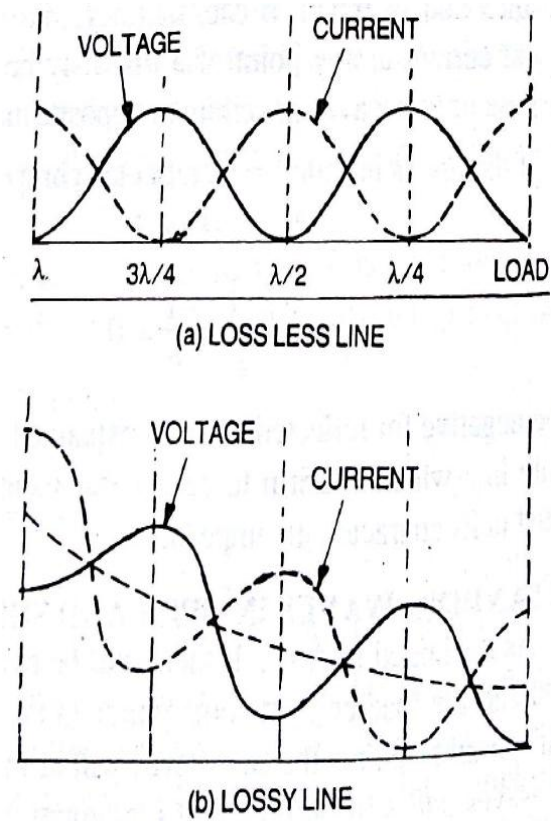


Fig. Voltage and current distribution on SC TL

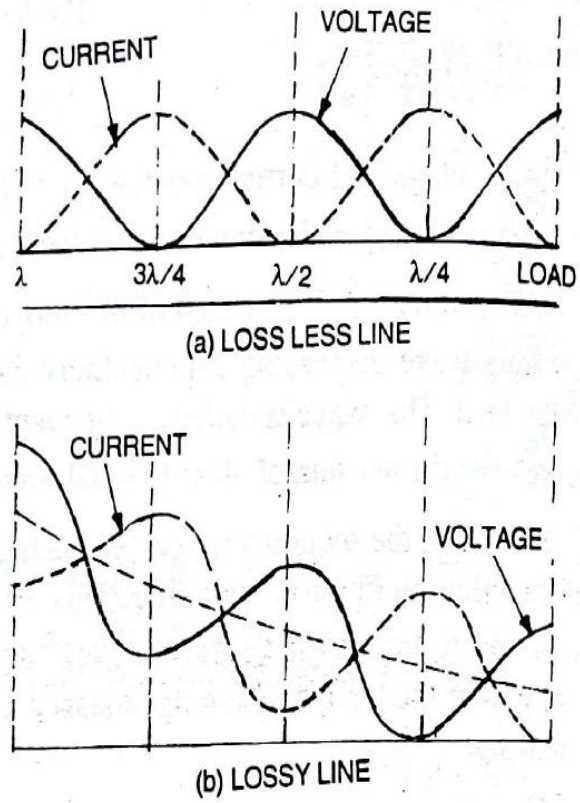


Fig. The variation of input impedance of OC line

<u>λ/8- line:</u> If length of the line is λ/8 then it is called λ/8 line or eight-wave line.

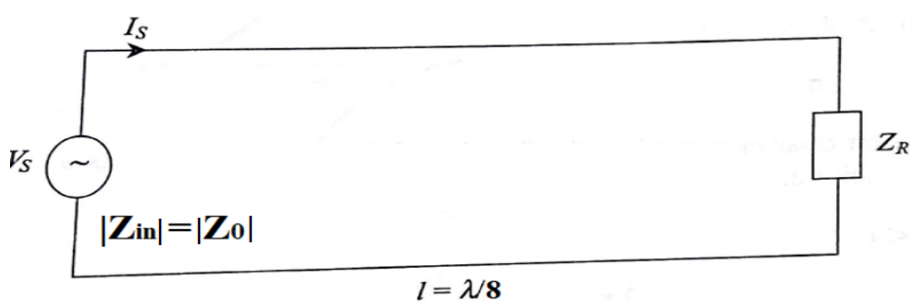


Fig. λ/8 line

The input impedance of the lossless transmission line is given by

$$Z_{in} = Z_0\left[\frac{Z_R + jZ_0 \tan\beta l}{Z_0 + jZ_R \tan\beta l}\right]$$

$$\beta l = \frac{2\pi}{\lambda}\frac{\lambda}{8} = \frac{\pi}{4}$$

$$Z_{in} = Z_0\left[\frac{Z_R + jZ_0 \tan\frac{\pi}{4}}{Z_0 + jZ_R \tan\frac{\pi}{4}}\right]$$

$$Z_{in} = Z_0\left[\frac{Z_R + jZ_0}{Z_0 + jZ_R}\right]$$

$$|Z_{in}| = |Z_0|$$

The λ/8 line is used to transform any load impedance Z_R to input impedance Z_{in} whose magnitude is equal to magnitude of Z_0.

<u>Quarter-wave line:</u> If length of the line is λ/4 then it is called λ/4 line or quarter-wave line.

The input impedance of the lossless transmission line is given by

$$Z_{in} = Z_0\left[\frac{Z_R + jZ_0 \tan\beta l}{Z_0 + jZ_R \tan\beta l}\right]$$

$$\beta l = \frac{2\pi}{\lambda}\frac{\lambda}{4} = \frac{\pi}{2}$$

$$Z_{in} = Z_0\left[\frac{Z_R + jZ_0 \tan\frac{\pi}{2}}{Z_0 + jZ_R \tan\frac{\pi}{2}}\right]$$

$$Z_{in} = Z_0\left[\frac{\frac{Z_R}{\tan\frac{\pi}{2}} + jZ_0}{\frac{Z_0}{\tan\frac{\pi}{2}} + jZ_R}\right]$$

$$Z_{in} = \frac{{Z_0}^2}{Z_R}$$

The λ/4 line can transform low impedance into high impedance and vice versa, thus it can be considered as an impedance inverter.

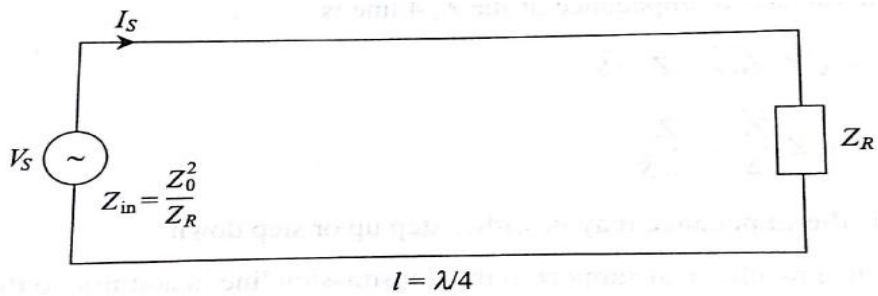


Fig. Quarter-wave line

The quarter wave line may be used as an impedance transformer for matching of load impedance Z_R with input impedance Z_{in}. For impedance matching Z_R and Z_{in} , the line characteristic impedance Z_o may be selected as $Z_o = \sqrt{Z_R Z_{in}}$

<u>λ/2- line(half-wave line):</u> If length of the line is λ/2 then it is called λ/2 line or half-wave line.

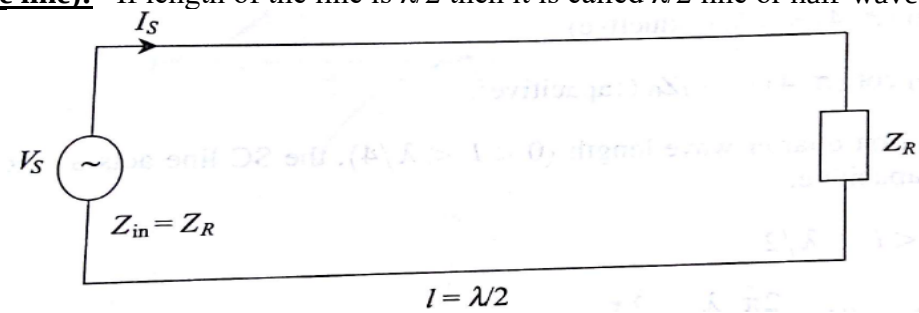


Fig. Half-wave line.

The input impedance of the lossless transmission line is given by

$$Z_{in} = Z_0\left[\frac{Z_R + jZ_0 \tan\beta l}{Z_0 + jZ_R \tan\beta l}\right]$$

$$\beta l = \frac{2\pi}{\lambda}\frac{\lambda}{2} = \pi$$

$$Z_{in} = Z_0\left[\frac{Z_R + jZ_0 \tan\pi}{Z_0 + jZ_R \tan\pi}\right]$$

$$Z_{in} = Z_R$$

The input impedance Z_{in} of the λ/2 line is equal to load impedance Z_R.

Transmission lines of various lengths can be used as circuit elements: Different lengths of transmission lines can be used as circuit elements as discussed below.

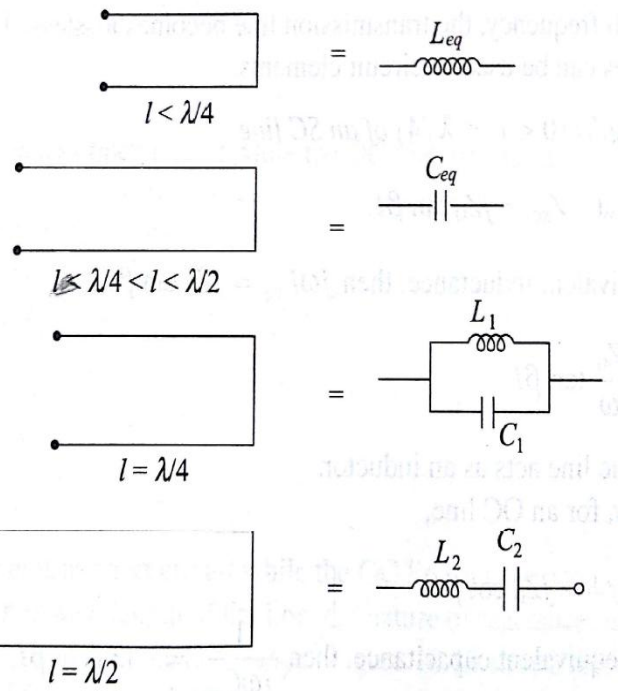


Fig. SC lines at different lengths with equivalent circuits

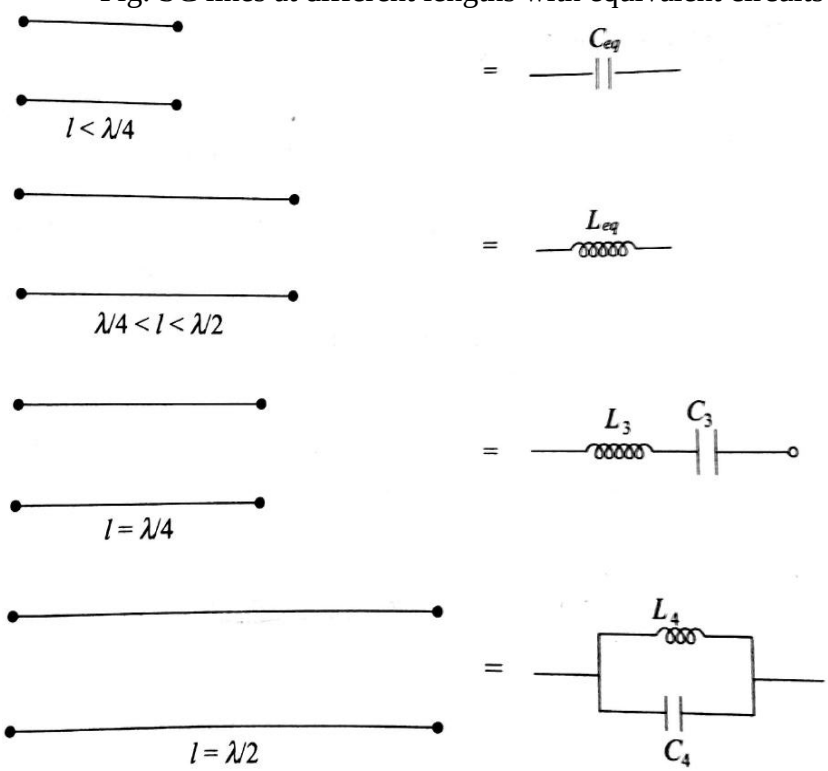


Fig. OC lines at different lengths with equivalent circuits

The input impedance of the short circuited lossless transmission line is given by

$$Z_{sc} = jZ_0 \tan \beta l \quad --- (1)$$

The input impedance of the open circuited lossless transmission line is given by

$$Z_{oc} = -j\, Z_0 \cot \beta l \quad --- (2)$$

The Eq.s(1) and (2) shows that input impedance of an open and short circuited lossless line is a pure reactance. Desired value of the reactance is obtained by varying the electrical length βl of the stubs.

If length of the short circuited line is less than λ/4, It will act as inductance. If length of the short circuited line is greater than λ/4 and less then λ/2, It will act as capacitance. If length of the short circuited line is equal

to λ/4, It will act as parallel resonance circuit with high impedance. If length of the short circuited line is equal to λ/2, It will act as series resonance circuit with low impedance.

If length of the open circuited line is less than λ/4, It will act as capacitance. If length of the open circuited line is greater than λ/4 and less then λ/2, It will act as inductance. If length of the open circuited line is equal to λ/4, It will act as series resonance circuit with low impedance. If length of the open circuited line is equal to λ/2, It will act as parallel resonance circuit with high impedance.

<u>The Smith Chart:</u>

The Smith chart is the most commonly used graphical techniques in solving transmission problems in simple way. It is basically a graphical indication of the impedance and VSWR of a transmission line as one moves along the line.

The construction of the chart is based on the reflection coefficient

$$K = \frac{Z_R - Z_0}{Z_R + Z_0}$$

Instead of having separate Smith charts for transmission lines with different characteristic impedances such as Zo = 60,100, and 120. A normalized chart can be used for any line in which all impedances are normalized with respect to the characteristic impedance Zo of the particular line under consideration.

$$K = \frac{\frac{Z_R}{Z_0} - 1}{\frac{Z_R}{Z_0} + 1} = \frac{z_r - 1}{z_r + 1}$$

Where z_r is the normalized impedance

$$z_r = \frac{1 + K}{1 - K}$$

Since z_r and K both are complex quantities, we have

$$R + jX = \frac{1 + K_r + jK_x}{1 - (K_r + jK_x)}$$

$$R + jX = \frac{(1 + K_r) + jK_x}{(1 - K_r) - jK_x}$$

Rationalizing on the right hand side, we get

$$R + jX = \frac{(1 + K_r) + jK_x}{(1 - K_r) - jK_x} \times \frac{(1 - K_r) + jK_x}{(1 - K_r) + jK_x}$$

$$R + jX = \frac{1 - K_r^2 - K_x^2 + 2jK_x}{(1 - K_r)^2 + K_x^2}$$

Equating real and imaginary parts on both sides, we get

$$R = \frac{1 - K_r^2 - K_x^2}{(1 - K_r)^2 + K_x^2} \quad ----(1)$$

$$X = \frac{2K_x}{(1 - K_r)^2 + K_x^2} \quad ---(2)$$

Eq.s(1) and (2) will yield two set of orthogonal circles when solved separately. Eq.(1) will results in family of circle called R-circle while Eq.(2) will results in family of circle called X-circle.

(i) The constant R-circle:

$$R\{1 + K_r^2 - 2K_r + K_x^2\} = 1 - K_r^2 - K_x^2$$

$$R + RK_r^2 - 2RK_r + RK_x^2 = 1 - K_r^2 - K_x^2$$

$$K_r^2(R + 1) + K_x^2(R + 1) - 2RK_r = 1 - R$$

$$K_r^2 + K_x^2 - \frac{2R}{1 + R}K_r = \frac{1 - R}{1 + R}$$

Adding $\frac{R^2}{(1+R)^2}$ on both sides, we get

$$K_r^2 - \frac{2R}{1+R}K_r + \frac{R^2}{(1+R)^2} + K_x^2 = \frac{1-R}{1+R} + \frac{R^2}{(1+R)^2}$$

$$\left(K_r - \frac{R}{1+R}\right)^2 + K_x^2 = \left(\frac{1}{1+R}\right)^2$$

This equation represents a family of circles on the reflection co-efficient plane. These circles are called constant –R circles with center $\left(\frac{R}{1+R}, 0\right)$ and radius $\frac{1}{1+R}$.

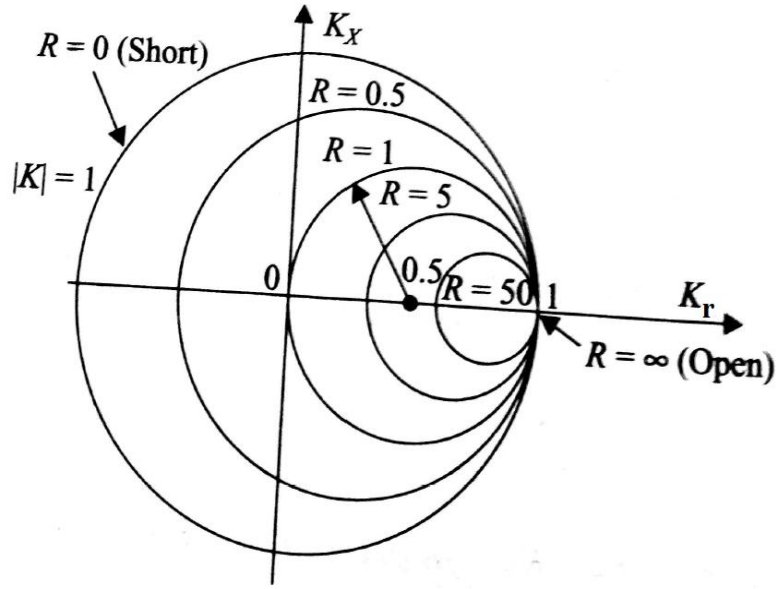


Fig. Family of constant R circles

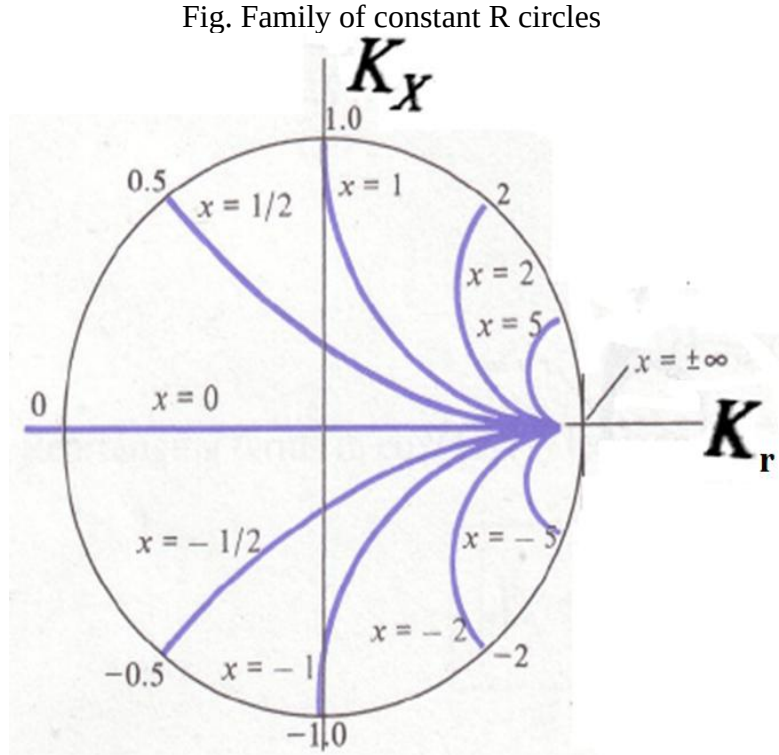


Fig. Family of constant X circles

(i) The constant X-circle:

$$X = \frac{2K_x}{(1-K_r)^2 + K_x^2}$$

$$(1-K_r)^2 + K_x^2 = \frac{2K_x}{X}$$

$$(1-K_r)^2 + K_x^2 - \frac{2K_x}{X} = 0$$

Adding $\left(\frac{1}{X}\right)^2$ on both sides, we get

$$(1-K_r)^2 + K_x^2 - \frac{2K_x}{X} + \left(\frac{1}{X}\right)^2 = \left(\frac{1}{X}\right)^2$$

$$(K_r-1)^2 + \left(K_x - \frac{1}{X}\right)^2 = \left(\frac{1}{X}\right)^2$$

This equation represents a family of circles on the reflection co-efficient plane. These circles are called constant –X circles with center $\left(1, \frac{1}{X}\right)$ and radius $\frac{1}{X}$.

The complete smith chart can be obtained by the superposition of the family of the constant-R circles and constant-X circles on K plane as shown in fig.

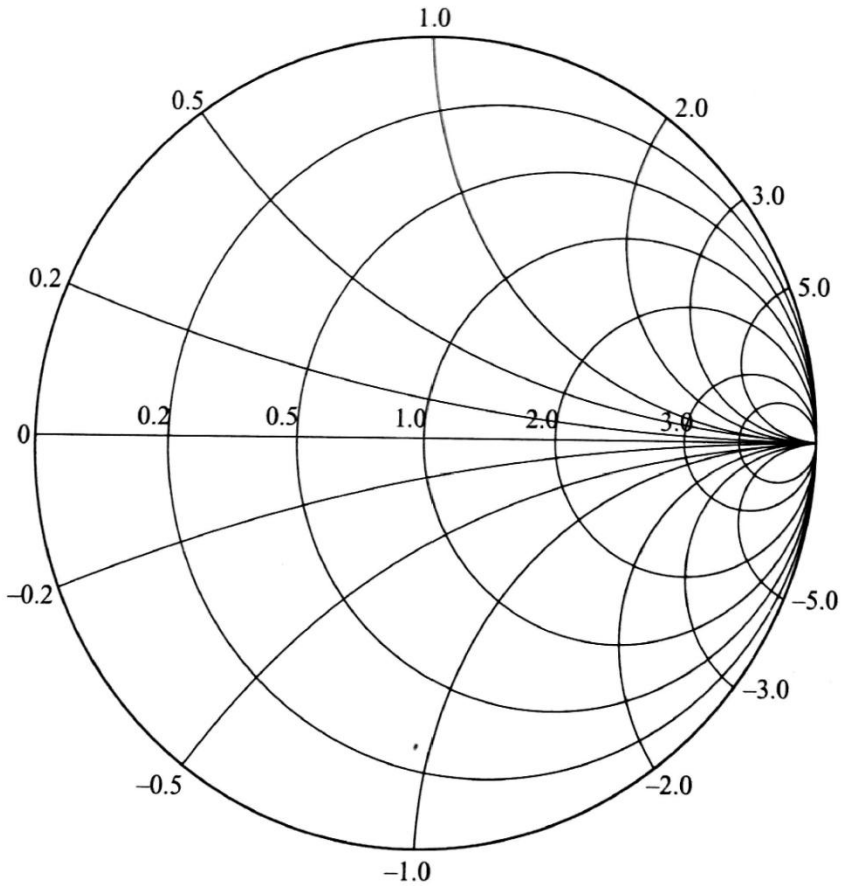


Fig. Smith chart

Properties of smith chart:

1. The smith chart can be used for impedance as well as for admittance.
2. The smith chart consists of constant –R circles and constant –X circles.
3. The point P_{sc} on the chart represents short circuit or zero impedance and the point P_{oc} on the chart represents open circuit or infinite impedance.
4. A complete revolution of 360° around the Smith chart represents a distance of λ/2 on the line.
5. Clockwise movement on the chart is regarded as moving toward the generator from the load on the transmission line. Similarly, counterclockwise movement on the chart corresponds to moving toward the load from the generator.
6. There are three scales around the periphery of the Smith chart as illustrated in Figure. The outermost scale is used to determine the distance on the line from the generator end in terms of wavelengths, and the next scale determines the distance from the load end in terms of wavelengths. The innermost scale is used to determine angle of reflection coefficient in degrees.
7. Since a λ /2 distance on the line corresponds to a movement of 360° on the chart, λ distance on the line corresponds to a 720° movement on the chart.
8. V_{max} occurs where $Z_{in(max)}$ located on the chart while V_{min} occurs where $Z_{in(min)}$ located

9. The distance between Z_{sc} and Z_{oc} is $\lambda/4$.

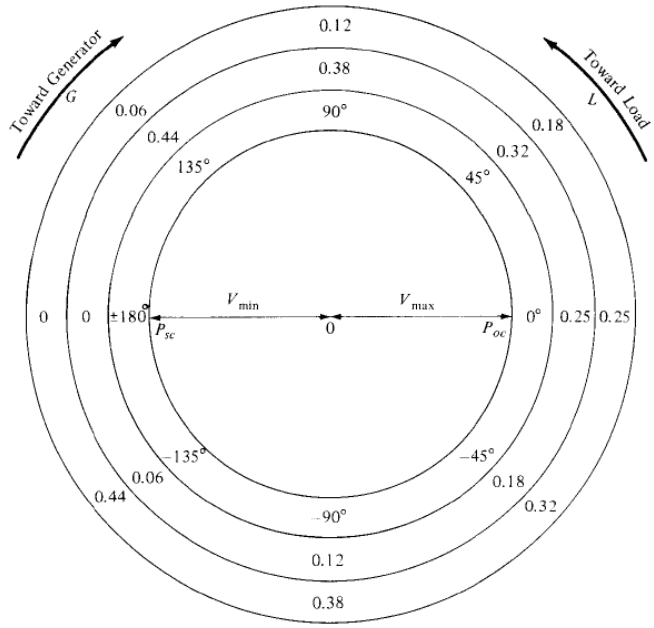


Fig. Smith chart illustrating scales around the periphery and movements around the chart line.

<u>Applications of smith chart:</u>

1. Plotting impedance
2. Measurement of VSWR
3. Measurement of reflection coefficient K
4. Measurement of input impedance of the transmission line.
5. Impedance to admittance conversion
6. Finding voltage maximum and voltage minimum locations
7. Stub matching

<u>Microstrip transmission line:</u>

A microstrip line consists of a single ground plane and an open strip conductor separated by dielectric substrate as shown in Figure.

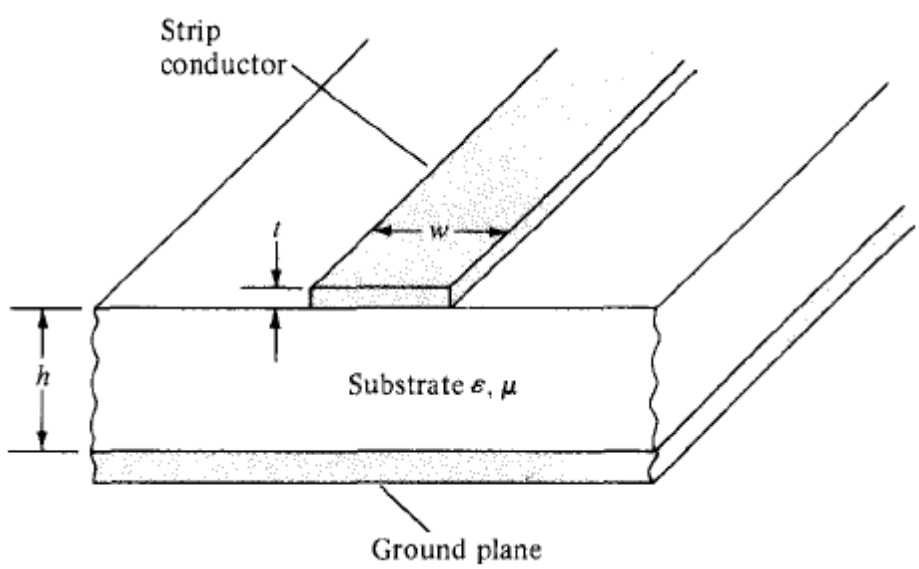


Fig. Microstrip line

The characteristic impedance of microstrip line is given by the following approximate formula

$$Z_o = \begin{cases} \dfrac{60}{\sqrt{\varepsilon_{\text{eff}}}} \ln\left(\dfrac{8h}{w} + \dfrac{w}{h}\right), & w/h \le 1 \\ \dfrac{1}{\sqrt{\varepsilon_{\text{eff}}}} \dfrac{120\pi}{[w/h + 1.393 + 0.667 \ln (w/h + 1.444)]}, & w/h \ge 1 \end{cases}$$

$$\varepsilon_{\text{eff}} = \frac{(\varepsilon_r + 1)}{2} + \frac{(\varepsilon_r - 1)}{2\sqrt{1 + 12h/w}}$$

The input impedance of the lossless microstrip line is given by

$$Z_{in} = Z_0 \left[\frac{Z_R + jZ_0 \tan \beta l}{Z_0 + jZ_R \tan \beta l}\right]$$

<u>Impedance matching techniques:</u>

When the transmission line is terminated with a load impedance which is not equal to the characteristic impedance of the line, mismatch occurs and reflected wave exists on the line. Mismatch reduces efficiency and increases power loss. To avoid mismatching, it is necessary to add impedance matching device between the load and the line.

The following are the impedance matching techniques

1. Quarter-wave impedance matching or quarter-wave transformer
2. Stub matching

<u>Quarter wave impedance matching or quarter wave transformer:</u>

When the transmission line is terminated with a load impedance which is not equal to the characteristic impedance of the line, mismatch occurs and reflected wave exists on the line. Mismatch reduces efficiency and increases power loss. Quarter wave line is inserted between the line and load to match load impedance to the line as shown in Fig.

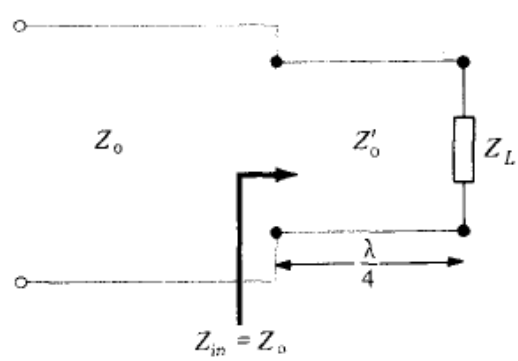


Fig. Quarter wave transformer

Z_0^1 is selected such that $Z_{in} = Z_0$

$$Z_0^l = \sqrt{Z_0 Z_L}$$

The λ/4 line is also called Quarter wave transformer.

Advantages:

1. Simple to design

Disadvantages:

1. We have to cut the line to insert a quarter wave transformer in between the line and the load.
2. It is frequency sensitive

Stub matching:

A piece of transmission line is called stub. The stub may be open or short circuited. A small section of open or short circuited stub is used as impedance matching device, which can be connected in parallel to the line at a certain distance from the load. This matching device is called stub matching.

The stub has the same characteristic impedance as the main line. It is more difficult to use a series stub although it is theoretically feasible. An open-circuited stub radiates some energy at high frequencies. Consequently, short-circuited parallel stubs are preferred.

There two types of stub matching methods

1. Single stub matching
2. Double stub matching

1. Single stub matching:

In this method, to achieve impedance matching, a short circuited stub is connected in parallel to the line at a certain distance from the load as shown in Fig. Since the stub is connected in parallel, it is easy to use admittance instead of impedance for analysis.

When load admittance Y_R is connected to the line and if it is not equal to the characteristic admittance Y_0, mismatch occurs and reflected wave exists on the line.

The input admittance at point 1 looking towards load is given by

$$Y_{in} = Y_0 \pm jB$$

This is the admittance at point 1 before stub is connected. The point 1 is selected such that at point 1, $Y_0=1/Z_0$. The short circuited stub is connected at point 1 in parallel with the main line. The length of the stub is selected such that its input succeptance is $\mp jB$.

The total input admittance at 1 is

$$Y_{in} = Y_0 \pm jB \mp jB = Y_0$$

Thus the input impedance looking towards load is

$$Z_{in} = Z_0$$

Therefore the line terminated with Z_0 at point 1, hence the line is said be matched.

Length and location stub:

Consider a transmission line terminated with load admittance Y_R. Let a short circuited stub of length l_t is connected to the main line at a distance l_s from the load as shown in Fig.

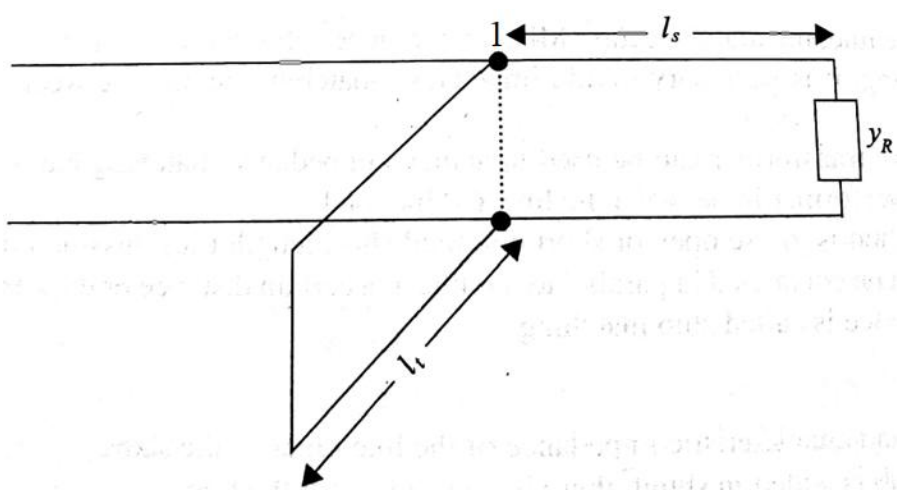


Fig. Single stub matching

We know that the input impedance at any point on the line is

$$Z_{in} = Z_0 \left[\frac{Z_R + jZ_0 \tan \beta l}{Z_0 + jZ_R \tan \beta l}\right]$$

The corresponding input admittance is given by

$$Y_{in} = \frac{1}{Z_{in}}$$

$$Y_{in} = Y_0 \left[\frac{Y_R + jY_0 \tan \beta l}{Y_0 + jY_R \tan \beta l}\right]$$

The normalized input admittance is given by

$$y_{in} = \frac{Y_{in}}{Y_0}$$

$$y_{in} = \left[\frac{\frac{Y_R}{Y_0} + j \tan \beta l}{1 + j\frac{Y_R}{Y_0} \tan \beta l}\right]$$

$$y_{in} = \left[\frac{y_R + j \tan \beta l}{1 + jy_R \tan \beta l}\right]$$

Where the normalized load admittance is

$$y_R = \frac{Y_R}{Y_0}$$

Separating the real and imaginary terms

$$y_{in} = \left(\frac{y_R + j \tan \beta l}{1 + jy_R \tan \beta l}\right)\left(\frac{1 - jy_R \tan \beta l}{1 - jy_R \tan \beta l}\right)$$

$$y_{in} = \frac{y_R + y_R \tan^2 \beta l + j \tan \beta l - jy_R^2 \tan \beta l}{1 + y_R^2 \tan^2 \beta l}$$

$$y_{in} = \frac{y_R(1 + \tan^2 \beta l)}{1 + y_R^2 \tan^2 \beta l} + j\frac{(1 - y_R^2) \tan \beta l}{1 + y_R^2 \tan^2 \beta l}$$

For no reflection, at distance $l = l_s$, the real part of normalized input impedance is equal to unity. Therefore

$$\frac{y_R(1 + \tan^2 \beta l_s)}{1 + y_R^2 \tan^2 \beta l_s} = 1$$

$$y_R(1 + \tan^2 \beta l_s) = 1 + y_R^2 \tan^2 \beta l_s$$

$$y_R + y_R \tan^2 \beta l_s = 1 + y_R^2 \tan^2 \beta l_s$$

$$(y_R - y_R^2) \tan^2 \beta l_s = 1 - y_R$$

$$\tan^2 \beta l_s = \frac{1 - y_R}{y_R(1 - y_R)}$$

$$\tan^2 \beta l_s = \frac{1}{y_R}$$

$$\tan \beta l_s = \frac{1}{\sqrt{y_R}} = \sqrt{\frac{Y_0}{Y_R}}$$

$$l_s = \frac{1}{\beta} \tan^{-1} \sqrt{\frac{Y_0}{Y_R}}$$

Therefore the location of stub is given by

$$l_s = \frac{\lambda}{2\pi} \tan^{-1} \sqrt{\frac{Z_R}{Z_0}}$$

At this location the imaginary part of y_{in} is

$$b_s = \frac{(1 - \mathrm{y}_\mathrm{R}^2)\tan\beta l_s}{1 + \mathrm{y}_\mathrm{R}^2 \tan^2\beta l_s}$$

But

$$\tan\beta l_s = \frac{1}{\sqrt{y_R}}$$

$$b_s = \frac{(1 - \mathrm{y}_\mathrm{R}^2)\frac{1}{\sqrt{y_R}}}{1 + \mathrm{y}_\mathrm{R}^2 \frac{1}{y_R}}$$

$$b_s = \frac{(1 - \mathrm{y}_\mathrm{R}^2)}{\sqrt{y_R}(1 + y_R)}$$

$$b_s = \frac{(1 - y_R)}{\sqrt{y_R}}$$

$$b_s = \frac{\left(1 - \frac{\mathrm{Y}_R}{\mathrm{Y}_0}\right)}{\sqrt{\frac{\mathrm{Y}_R}{\mathrm{Y}_0}}}$$

$$b_s = \frac{(\mathrm{Y}_0 - \mathrm{Y}_R)}{\sqrt{\mathrm{Y}_R\mathrm{Y}_0}}$$

The input impedance of the short circuited stub is given by

$$Z_{sc} = j\mathrm{Z}_0 \tan\beta l_t$$

The normalized input admittance is

$$y_{sc} = -j\cot\beta l_t$$

The normalized succeptance of stub is

$$b_{ss} = -\cot\beta l_t$$

At point 1 the total succeptance is zero

$$b_s + b_{ss} = 0$$

$$\frac{(\mathrm{Y}_0 - \mathrm{Y}_R)}{\sqrt{\mathrm{Y}_R\mathrm{Y}_0}} - \cot\beta l_t = 0$$

$$\cot\beta l_t = \frac{(\mathrm{Y}_0 - \mathrm{Y}_R)}{\sqrt{\mathrm{Y}_R\mathrm{Y}_0}}$$

$$\tan\beta l_t = \frac{\sqrt{\mathrm{Y}_R\mathrm{Y}_0}}{(\mathrm{Y}_0 - \mathrm{Y}_R)}$$

$$\tan\beta l_t = \frac{\sqrt{Z_R\mathrm{Z}_0}}{(\mathrm{Z}_\mathrm{R} - Z_0)}$$

$$l_t = \frac{1}{\beta}\tan^{-1}\left(\frac{\sqrt{Z_R\mathrm{Z}_0}}{(\mathrm{Z}_\mathrm{R} - Z_0)}\right)$$

Therefore the length of the stub is

$$l_t = \frac{\lambda}{2\pi}\tan^{-1}\left(\frac{\sqrt{Z_R\mathrm{Z}_0}}{(\mathrm{Z}_\mathrm{R} - Z_0)}\right)$$

<u>Disadvantages of single stub matching:</u>

1. The location and length of the stub are frequency dependent. If frequency of the wave changes, the location and length of the stub should be changed.
2. In practical cases, the location of stub has to be moved along the line for fine adjustment.

<u>Double stub matching:</u>

To overcome the drawbacks of the single stub matching, two stubs are used at different locations. This is called double stub matching.
Consider a double stub matching system consisting of two short circuited stubs connected in parallel to the main line near the load as shown in Fig.

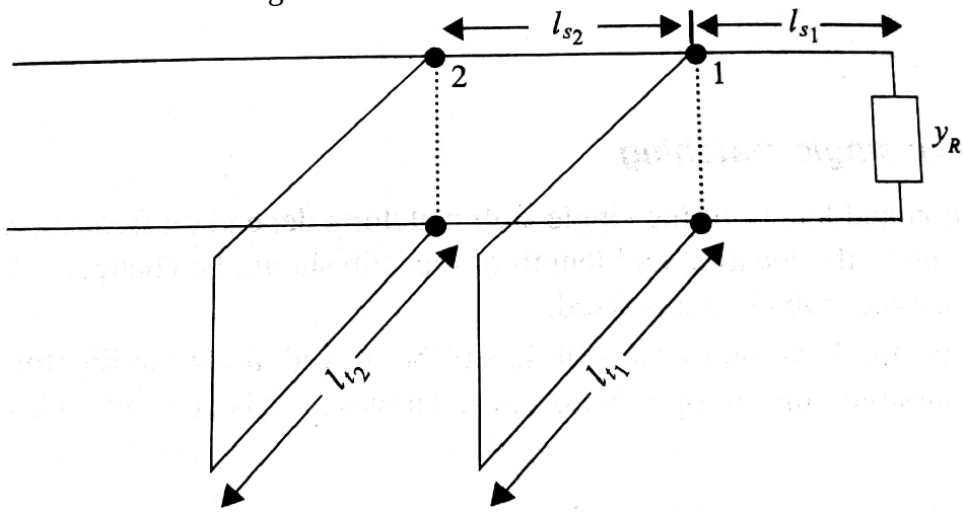


Fig. double stub matching

The characteristic admittance of the stubs should be equal to the characteristic admittance of the line. In double stub matching the locations of the stubs are arbitrary, but the spacing between two stubs must be equal to λ/8, λ/4 or 3λ/8.
Therefore, in the design of double stub matching, it is better to keep the locations of stubs fixed. Impedance matching is done by finding the lengths of the stubs. When the frequency changes, the stub lengths can be adjusted to achieve impedance matching.

Design of double stub matching:

1. The normalized admittance at the location of stub1 from load is

$$y_A = g_A + jb_A$$

2. Connect a short circuited stub1 having succepatance $\pm jb_1$ to the line at point 1
The normalized admittance at point 1 after stub connection is

$$y_A = g_A + j(b_A \pm b_1)$$

3. For the impedance matching, the normalized admittance at the location of stub2 is

$$y_B = 1 \pm jb_B$$

4. The spacing between two stubs is selected such the real part of normalized admittance at the location of stub2 is unity.
5. Connect second short circuited stub having succeptance of $\mp jb_B$to the line at 2
Now the normalized input admittance at point 2 after stub2 connection

$$y_{in} = 1 \pm jb_B \mp jb_B = 1$$

$$y_{in} = 1$$

$$Z_{in} = Z_0$$

Therefore the line terminated with Z_0 at point 2, hence the line is said be matched.

Loading:

The processes of increasing the inductance L of the transmission line artificially is called loading. And such a line is called loaded line.
There are three types of loading methods
1. Patched loading
2. Continuous loading
3. Lumped loading